Problem #5

Due: Friday, December 5, turn in to canvas, as described in the syllabus.

1. Please do and write up solutions to the following three problems from the textbook. (Note: pictures of the problems from Meiss are appended to this assignment.)

   (a) Page 187, Section 5.7: Problem 10

   (b) Page 314-316, Section 8.15, Problems 8

   (c) Page 314-316, Section 8.15: Problem 13

2. In a 1999 Science paper, Klausmeier proposed a simple model to explain properties of vegetation patterns that have been found in drylands of Africa, Australia and the Americas. Here is a google map link if you want to explore: https://maps.app.goo.gl/rfKWVC7WuqNWPFKw8.

   (In this map, the orange is bare soil and the black is the vegetation. Note that the vegetation is concentrated in bands that are fairly regularly spaced. They are oriented approximately perpendicular to a gentle elevation grade. It turns out that these vegetation bands also slowly migrating upslope, via upslope colonization of the plants, over the course of decades.)

   The nondimensionalized version of his model (in one space dimension) is

$$
\begin{aligned}
\frac{\partial w}{\partial t} &= p - w - wb^2 + \nu \frac{\partial w}{\partial x} \\
\frac{\partial b}{\partial t} &= -mb + wb^2 + \frac{\partial^2 b}{\partial x^2}.
\end{aligned}
$$

   - Here $w(x,t) \geq 0$ represents available water and $b(x,t) \geq 0$ represents the biomass density.
   - The parameter $p \geq 0$ represents a constant (mean annual) precipitation input to the system.
   - The parameter ν is controlled by the slope of the ground in the x-direction, which leads to downhill runoff of the water. We'll assume that ν is constant, and that its value increases with the (constant) slope of the ground in the x-direction.
   - The term $-w$ in the water equation models evaporative losses.
   - The nonlinear growth term wb^2 in the biomass equation assumes that the biomass growth rate increases with both the water w and the amount of biomass b. (The nonlinear dependence on b captures positive feedbacks between water availability and biomass presence. This is due to infiltration feedbacks that concentrate the water resource where it is needed. These plants have been referred to as 'ecosystem engineers'.) Since this growth is a result of transpiration, there is a corresponding loss term $-wb^2$ in the water equation.
   - The parameter $m > 0$ represents the natural death rate for the plants.
   - Seed dispersal is modeled by the diffusion term in the biomass equation.

   This problem is concerned with bifurcations of the model with changes in mean precipitation p, which we treat as a bifurcation parameter.

(a) Does $\nu > 0$ describe a hill with higher elevation to the left or to the right (where x increases as you move from left to right)? Briefly explain your reasoning.

(b) Consider the mortality m to be a fixed positive number, and construct a qualitative bifurcation diagram that shows all spatially uniform, stationary solutions as a function of the mean annual precipitation level p (i.e. solutions with $\frac{\partial b}{\partial t} = \frac{\partial b}{\partial x} = 0$, etc.). For this bifurcation diagram, you should plot the biomass level b as a function of the precipitation level p qualitatively, and identify any precipitation threshold levels of p_c where there is a change in the number of stationary, uniform solutions. Include the stability of the solutions within this framework. Indicating stable solutions as solid lines and unstable ones as dashed lines.

(c) We can convert the partial differential equations describing interactions between biomass and water to a system of ordinary differential equations in a traveling frame by introducing a variable $z = x - ct$ that will allow us to capture traveling wave solutions. This leads to a first order differential equation for w and a second order differential equation for b. Rewrite the differential equations in the z coordinate as a system of three ordinary differential equations by introducing a new variable $u = db/dz$.

(d) The equilibrium solutions of the system of ODEs for $(b(z), u(z), w(z))$ are identical to those determined in your answer to question (2). It is possible to show that the equilibrium solution with the largest value of biomass b can undergo a Hopf bifurcation with a linear frequency ω under certain conditions on p and c. Describe how you would determine the conditions for a Hopf bifurcation and how you would interpret a periodic solution branch born from this bifurcation in the original setting of the vegetation model. What do the values of p and c indicate? How about the frequency ω? Here, I am not necessarily asking you to do the calculations, but for full credit you must carefully and accurately outline how you'd do them, and, importantly, how you would interpret your results, especially in light of what I've told you about these ecosystems.

Section 5.7: Problem 10

10. The three-dimensional system

$$\begin{aligned} \dot{x} &= y + 2z + (x+z)^2 + xy - y^2, \\ \dot{y} &= (x+z)^2, \\ \dot{z} &= -2z - (x+z)^2 + y^2 \end{aligned} \tag{5.43}$$

has a nonhyperbolic equilibrium at the origin.

(a) Find a linear transformation to write (5.43) in the form (5.34).

(b) Find the quadratic approximation for $W^c(0,0,0)$.

(c) Obtain the reduced dynamics (5.36) on W^c and use your favorite software package to study it. Is the origin stable or unstable?

Section 8.15: Problems 8

8. Verify the calculations leading to the normal form (8.43) of the center in $\mathbb{R}^2$. In particular derive the homological operator L_A, find its action on the standard bases of $\mathbb{H}_2^2$ and $\mathbb{H}_3^2$, and obtain the matrices L. Find the eigenvectors and eigenvalues. Show that the null, left eigenvectors in $\mathbb{H}_3^2$ are given by (8.42) but that the null, right eigenvectors, together with range of $L_A(\mathbb{H}_3^2)$, do indeed span $\mathbb{H}_3^2$.

Section 8.15: Problem 13

Determine whether it is subcritical or supercritical.

13. Consider the system

$$\begin{aligned} \dot{x} &= \mu x - y + ay^2 + x^3, \\ \dot{y} &= x + \mu y + xy^2 + y^2. \end{aligned}$$

(a) Determine $\alpha(a)$ using (8.59).

(b) Find the set on which this system has an Andronov–Hopf bifurcation. Is the bifurcation subcritical or supercritical?

(c) Investigate numerically the behavior for values of a such that $\alpha > 0$, $\alpha < 0$, and $\alpha = 0$.